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RESEARCH

Risk of cardiovascular events according to the tricyclic antidepressant dosage in patients with chronic pain: a retrospective cohort study

Hyunji Koo[1] · Seung Hun You[2] · Sewon Park[2] · Kyeong Hye Jeong[1] · Nakyung Jeon[3] · Sun-Young Jung[1,2]

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Abstract

Purpose We aimed to examine the risk of cardiovascular adverse events by tricyclic antidepressant (TCA) dosage among patients with chronic pain.

Methods A retrospective cohort study was conducted using a nationwide sample cohort. Among patients aged ≥ 18 years with a chronic pain diagnosis and no history of cardiovascular events, we extracted users and non-users of TCAs through 1:1 propensity score matching. TCA users were categorized into three groups according to the mean defined daily dose (DDD): very low doses (< 0.15 DDD), low doses (0.15–0.34 DDD), and traditional doses (≥ 0.34 DDD). A 6-month follow-up was conducted with an intention-to-treat approach. We examined the hazard ratio of cardiovascular adverse events using Cox proportional hazards analysis.

Results In total, 16,660 matched patients were followed up (8330 TCA users and 8330 non-users). TCA use did not significantly increase cardiovascular adverse events (hazard ratio [HR] 1.12, 95% confidence interval [CI] 0.94–1.33). Low-dose (0.15–0.34 DDD) TCAs (HR 1.37, 95% CI 1.08–1.74), particularly low-dose (0.15–0.34 DDD) nortriptyline (HR 2.11, 95% CI 1.44–3.08), was associated with an increased risk of cardiovascular adverse events. Administration of TCAs at the traditional dose (≥ 0.34 DDD) increased the risk of ischemic stroke (HR 2.08, 95% CI 1.11–3.88).

Conclusion Close monitoring of patients on long-term, low-dose use of TCAs should be conducted to avoid an increase in the cumulative dose, which increases the risk of cardiovascular adverse events.

Keywords Tricyclic antidepressants · Chronic pain · Cardiovascular adverse events · Low dose

Introduction

Chronic pain is a highly prevalent condition that impairs the quality of daily life [1]. It is defined as pain that persists or recurs for more than 3 months and requires continuous treatment and care. Headache, neuropathic pain, lower back pain, and fibromyalgia are well-known chronic pain conditions [1, 2]. Tricyclic antidepressants (TCAs) are one of the options used for chronic pain treatment, especially neuropathic pain [1, 3], and as first-line prophylactic treatment for tension-type headache [4]. Until recently, studies have continued to consider TCAs as a treatment method for various types of pain including lower back [5], neck [6], and chest [7] pains. The main mechanism underlying the analgesic effects of TCAs has been identified as the reinforcement of the descending inhibitory pathways by increasing norepinephrine and serotonin concentrations in the synaptic cleft at both the supraspinal and spinal levels [8].

TCAs have been associated with QT prolongation and sudden cardiac death [9, 10]. Moreover, a number of recent studies have shown that antidepressants, especially TCAs, can increase cardiovascular adverse events, such as myocardial infarction, stroke, arrhythmia, and cardiovascular death [11–14]. TCAs are known to have anticholinergic effects, such as orthostatic hypotension and increased heart rate, which can cause adverse cardiovascular events [15, 16]. A study on patients with depression showed that TCAs were associated with an increased risk of major adverse

✉ Sun-Young Jung
jsyoung@cau.ac.kr

1 College of Pharmacy, Chung-Ang University, 84 Heukseok-Ro, Dongjak-Gu, Seoul 06974, Republic of Korea

2 Department of Global Innovative Drugs, Graduate School of Chung-Ang University, Chung-Ang University, Seoul, South Korea

3 College of Pharmacy, Pusan National University College of Pharmacy, Pusan, South Korea

cardiovascular events (hazard ratio [HR] 1.2, 95% confidence interval [CI] 1.03–1.40) [13]; moreover, another study showed that TCAs were related to an increased risk of out-of-hospital cardiac arrest (OR 1.69, 95% CI 1.14–2.50) [12].

TCAs are known to provide a therapeutic effect against chronic pain, but to reduce adverse events, such as cardiovascular complications, lower doses than those used for depression are recommended [17]: for low back pain (LBP), 25 mg/day of amitriptyline [5]; for chronic neck pain, 5 mg/day of amitriptyline [6]; for chest pain, 10 mg/day of amitriptyline [7]; and for multiple functional somatic syndromes, 25–75 mg/day of imipramine [18]. Although the dosages administered to patients with chronic pain and depression can vary, few studies have investigated the association between dosage and the occurrence of cardiovascular adverse events in patients with chronic pain. Our key question was whether the use of low-dose TCAs is safe in terms of the risk of cardiovascular adverse events. Therefore, we examined the risk of cardiovascular adverse events in patients with chronic pain by considering the dose and type of TCA administered.

Methods

Study design and data source

We conducted a retrospective cohort study using the nationwide cohort database of the South Korea National Health Insurance Service–National Sample Cohort (NHIS-NSC) (Data Number: NHIS-2020–2-092). The NHIS-NSC database was compiled by extracting a nationwide sample consisting of 1,000,000 people from the Korean population in 2006, which is approximately 2.1% of the entire South Korean population. The NHIS-NSC database consists of anonymized personal identification information on sex, medical care history, medical care institution, diagnosis code, socioeconomic status, and drug prescription data collected from 2002 to 2015. This study protocol was exempted from review by the Institutional Review Board of Chung-Ang University (IRB Number: 1041078–202,002-HR-022–01).

Study population

From the entire database, we were able to identify patients with chronic pain as those diagnosed with the following the International Statistical Classification of Diseases and Related Health Problems 10th Revision (ICD-10) codes: headache (ICD-10: G43, G44, and R51) [19]; LBP (ICD-10: S33, S335, S336, S337, M544, M545, M541, M51, M510, M511, M512, M513, M514, M518, M519, M480, M993, M994, M995, M996, and M997) [20]; and diabetic neuropathic pain (DNP) (ICD-10: E104, E114, E124, E134, E144, G592, G632, and G99) [21]. Other chronic pain conditions included neck pain; non-inflammatory arthritic disorders; central pain syndrome; fibromyalgia; orofacial, ear, and temporomandibular pain disorders; abdominal and bowel pain; urogenital, pelvic, and menstrual pain; and musculoskeletal chest pain [22–25]. The ICD-10 codes are listed in Supplementary Table S1. Among these, we selected only those patients who were newly diagnosed with chronic pain between 2005 and 2014. Those diagnosed with chronic pain from 2002 to 2004 or in 2015 were excluded to reduce the prevalence bias, and 2015 was excluded to secure a follow-up period of 1 year. Furthermore, the following patients were excluded from our study: (1) those prescribed TCAs prior to the time of diagnosis of chronic pain; (2) those prescribed TCAs for diseases other than chronic pain; (3) those with a history of cardiovascular outcomes prior to cohort entry; and (4) those aged < 18 years at the time of cohort entry (Fig. 1). To adjust for the baseline differences in risk factors for cardiovascular adverse events between TCA users and non-users, we extracted TCA users and non-users at a matching ratio of 1:1.

Exposure assessment

The TCAs determined as the drugs of interest in this study are as follows: amitriptyline (World Health Organization Anatomical Therapeutic Chemical [WHO ATC] code, N06AA09); clomipramine (WHO ATC code, N06AA04); doxepin (WHO ATC code, N06AA12); imipramine (WHO ATC code, N06AA02); and nortriptyline (WHO ATC code, N06AA10). An intention-to-treat approach was implemented because these drugs are often used repeatedly within a short prescription period, such as on an "as needed" basis [26]. We calculated the mean defined daily dose (DDD) of TCAs, as described subsequently. For a drug administered for its primary indication in adults, using the ATC/DDD system, the mean dose of TCAs was calculated as the total dose of TCAs prescribed during the study period divided by the total prescribed period, which was then divided by the average maintenance dose per day [27]. We categorized TCA users into three groups according to the mean DDD as follows: very low doses (< 0.15 DDD), low doses (0.15–0.34 DDD), and traditional doses (≥ 0.34 DDD). The TCA grouping according to the DDD was based on previous studies on amitriptyline treatment for headache or neuropathic pain that used very low, low, and traditional dose categories according to the initial and maintenance doses of the drug [28, 29].

Potential confounders

Potential confounders included demographics, lifestyle, comorbidities, and concomitant medication. Demographic characteristics, such as age, sex, economic status, geographic

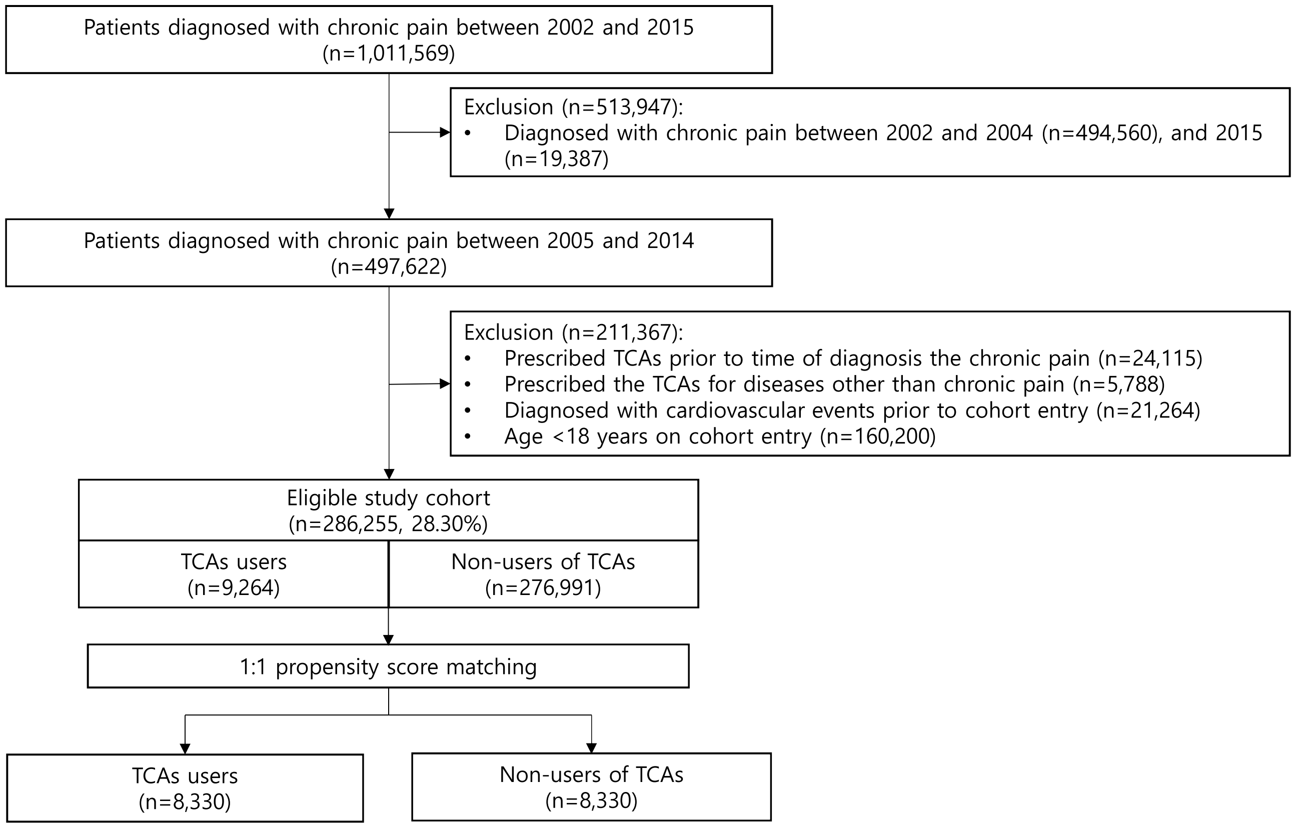


Fig. 1 Flow chart of study cohort formation. TCAs, tricyclic antidepressants

region (Seoul, metropolitan, and rural areas), inpatient status, type of medical institution, and physician specialization, were assessed on cohort entry. Lifestyle information (smoking habits, alcohol use, exercise habits, and body mass index) was assessed in the year prior to cohort entry. For comorbidities, the Charlson Comorbidity Index [30] (myocardial infarction, congestive heart failure, peripheral vascular disease, cerebrovascular disease, dementia, chronic pulmonary disease, rheumatoid arthritis/collagen vascular diseases, peptic ulcer disease, diabetes with/without chronic complications, hemiplegia or paraplegia, renal disease, and any malignancies [including lymphoma and leukemia, except malignant neoplasms of the skin]), hypertension, and psychiatric disease were assessed in the year prior to cohort entry. Headache, LBP, and DNP were identified as types of pain on cohort entry, and the number of emergency department visits, hospitalizations, and surgeries in the year prior to cohort entry were considered indications of pain severity. Furthermore, medication history (antiplatelets, statins, anticoagulants, non-steroidal anti-inflammatory drugs (NSAIDs), opioids, acetaminophen, tramadol, triptans, and anticonvulsants) was assessed in the year prior to cohort entry.

Outcomes and follow-up

Our primary outcome was a composite endpoint consisting of arrhythmia (ICD-10: I47, I48, I49), ischemic stroke (ICD-10: I63, I64, G45), hemorrhagic stroke (ICD-10: I60, I61, I62), heart failure (ICD-10: I50, I13, I109, I110), and myocardial infarction (ICD-10: I21, I22, I23, I24) [13, 14, 31, 32]. For our secondary outcomes, we separately assessed each of the following cardiovascular adverse events: arrhythmia, ischemic stroke, hemorrhagic stroke, heart failure, and myocardial infarction. We followed the patients included in the final cohort from the cohort entry date until the earliest of the following events: date of the cardiovascular outcome, date of censoring due to death, end of the study period (December 31, 2015), or the maximum follow-up of 180 days [5, 33, 34].

Statistical analyses

Patient baseline characteristics were evaluated using descriptive statistics, including frequencies, proportions, means, and standard deviations. To minimize potential confounders, propensity score matching using greedy 1:1 matching with a caliper of width 0.2 with exact matching for gender and cohort entry year was applied, and matching was performed using the PSMATCH procedure in SAS software (version 9.4; SAS, Institute, Inc., Cary, NC, USA). Patients were matched for all the characteristics shown in Table 1 including the cohort entry year. The standardized differences between TCA users and non-users were used to test for covariate imbalance before and after propensity score matching. A standardized difference < 0.1 indicates a well-balanced covariate. In our primary and secondary analyses, a Cox proportional hazards model was used to estimate the HR and 95% CI to compare cardiovascular events between TCA users and non-users. The proportionality of hazards assumption was assessed by the use of the Kolmogorov-type supremum test derived from cumulative sums of martingale residuals [35].

We then performed various subgroup analyses by stratifying by age group, sex, comorbidities (psychiatric disease, hypertension, diabetes mellitus), and concomitant medications. Concomitant medications were assessed during the follow-up period. Furthermore, we analyzed the impact of the association of the cumulative dose of TCAs with the outcome of interest. We calculated the cumulative prescribed period as the total sum of the TCA user's prescription duration and the cumulative DDD as the total sum of the DDD. All statistical analyses were performed using SAS software.

Results

Of the 1,011,569 patients who were diagnosed with chronic pain, we identified 497,622 patients who were newly diagnosed with chronic pain between 2005 and 2014. Of these, we excluded 24,115 who had been prescribed TCAs before their chronic pain diagnosis; 5788 patients who were prescribed TCAs for diseases other than chronic pain; 21,264 patients with a history of outcomes before cohort entry; and 160,200 patients aged < 18 years (Fig. 1). The remaining 286,255 patients were eligible for inclusion and consisted of 9264 TCA users and 276,991 non-users. TCA users (before matching) had an older mean age (44.37 vs. 37.95 years) and generally had more comorbidities (psychiatric disease, hypertension, and diabetic mellitus) and concomitant medications than non-users (Supplementary Table S2). Figure 2 shows the covariate balance before and after propensity score matching. After propensity score matching, 8330 TCA users were matched with 8330 non-users. The abovementioned differences among the cohorts before matching were reduced. Thus, the mean age of TCA users was 42.78 years, whereas that of non-users was 43.97 years (Table 1). Among TCA users, 6473 patients (77.71%) were prescribed amitriptyline, 1569 patients (18.84%) were prescribed nortriptyline, 262 patients (3.15%) were prescribed imipramine, and 26 patients (0.31%) were prescribed other TCAs or prescribed in duplicate. More than 60% of TCAs were prescribed at a very low dose (< 0.15 DDD) (Table 2). For the composite primary endpoint, 272 TCA users and 244 non-users experienced cardiovascular adverse events (Table 3). TCA users did not experience a significantly higher occurrence of cardiovascular adverse events than non-users (HR 1.12, 95% CI 0.94–1.33). However, when the TCA users were categorized according to the mean DDD, low-dose TCA users (0.15–0.34 DDD) exhibited a significantly increased risk of cardiovascular adverse events (HR 1.37, 95% CI 1.08–1.74). According to the type of TCA, the risk of cardiovascular adverse events was noted to have increased in the group administered low-dose (0.15–0.34 DDD) nortriptyline (HR 2.11, 95% CI 1.44–3.08) compared with non-users. No significant difference was found in terms of the risk of cardiovascular adverse events between each TCA drug group and non-users (Supplementary Table 3).

With regard to the type of cardiovascular adverse events, TCA users prescribing the traditional dose (≥ 0.34 DDD) showed an increased risk of ischemic stroke (HR 2.08, 95% CI 1.11–3.88) (Fig. 3). In the subgroup analysis, female patients (HR 1.35, 95% CI 1.05–1.75) showed a significantly increased risk of cardiovascular adverse events (Fig. 4). TCA users taking concomitant NSAIDs (HR 1.23, 95% 0.98–1.53) showed a marginally increased risk of cardiovascular adverse events. No significant differences were found in the risk of cardiovascular adverse events between TCA users and non-users among patients with headache, DNP, and LBP (Supplementary Table S4).

Additional analyses on the association between the cumulative dose or prescription duration of TCAs and cardiovascular adverse events confirmed that the hazard risk of adverse cardiovascular events was increased in users receiving > 3 DDD (HR 1.50, 95% CI 1.16–1.93) and a cumulative prescription duration of 8–30 days (HR 1.33, 95% CI 1.04–1.70) or 31–180 days (HR 1.53, 95% CI 1.11–2.12) (Supplementary Table S5).

Discussion

Our observational study evaluated the risk of cardiovascular adverse events in patients with chronic pain prescribed TCAs. In the six months of the intention-to-treat study, no significant difference was found in terms of the risk of cardiovascular events between matched TCA users and non-users. However, the results confirmed that the risk of cardiovascular adverse events increased in patients administered low-dose (0.15–0.34 DDD) TCAs, in patients with a cumulative dose of > 3 DDD, and in patients with a cumulative prescription duration of more than eight days.

Table 1 Baseline characteristics of TCA users and non-users after propensity-score matching

Characteristics	Non-users ($n = 8330$)	TCA users ($n = 8330$)	STD
Age, years			
Mean (±SD)	42.78 (± 14.90)	43.97 (± 14.80)	0.0128
18–39	3461 (41.55)	3477 (41.74)	
40–64	4073 (48.90)	3984 (47.83)	
65–79	715 (8.58)	773 (9.28)	
≥ 80	81 (0.97)	96 (1.15)	
Sex			0
Male	3920 (47.06)	3920 (47.06)	
Type of insurance			0.0308
Medical insurance	7628 (91.57)	7555 (90.70)	
Medical aid	702 (8.43)	775 (9.30)	
Inpatients	582 (6.99)	629 (7.55)	− 0.0217
Smoking habits[a]			0.0045
Non-smoker	2094 (25.14)	2042 (24.51)	
Ex-smoker	364 (4.37)	398 (4.78)	
Smoker	915 (10.98)	922 (11.07)	
Not available	4957 (59.51)	4968 (59.64)	
Alcohol habits[a]			0.0078
Non-drinker	2546 (30.56)	2465 (29.59)	
Trivial (2–3 days/month)	38 (0.46)	48 (0.58)	
Light (1–2 days/week)	2020 (24.25)	2105 (25.27)	
Moderate (3–4 days/week)	865 (10.38)	884 (10.61)	
Heavy (almost every day)	490 (5.88)	432 (5.19)	
Not available	2371 (28.46)	2396 (28.76)	
Exercise habits[a]			− 0.0068
None	680 (8.16)	679 (8.15)	
1–2 times/week	546 (6.55)	570 (6.84)	
3–4 times/week	374 (4.49)	361 (4.33)	
5–6 times/week	185 (2.22)	217 (2.61)	
Almost every day	205 (2.46)	186 (2.23)	
Not available	6340 (76.11)	6317 (75.83)	
Body mass index[a]			− 0.0058
Underweight	81 (0.97)	85 (1.02)	
Normal	838 (10.06)	820 (9.84)	
Overweight	494 (5.93)	524 (6.29)	
Obese 1	553 (6.64)	561 (6.73)	
Obese 2	68 (0.82)	69 (0.83)	
Not available	6296 (75.58)	6271 (75.28)	
Comorbidities[a]			
Myocardial infarction	0 (0.00)	3 (0.04)	0.0268
Congestive heart failure	11 (0.13)	9 (0.11)	− 0.0069
Peripheral vascular disease	387 (4.65)	425 (5.10)	0.0212
Cerebrovascular disease	159 (1.91)	186 (2.23)	0.0228
Dementia	10 (0.12)	7 (0.08)	− 0.0113
Chronic pulmonary disease	1290 (15.49)	1423 (17.08)	0.0433
Rheumatoid arthritis[b]	266 (3.19)	304 (3.65)	0.0251
Peptic ulcer disease	1550 (18.61)	1622 (19.47)	0.022
Mild liver disease	1010 (12.12)	1088 (13.06)	0.0282
Diabetic without chronic complication	548 (6.58)	619 (7.43)	0.0334
Diabetes with chronic complication	266 (3.19)	323 (3.88)	0.0371
Hemiplegia or paraplegia	18 (0.22)	26 (0.31)	0.0187

Table 1 (continued)

Characteristics	Non-users ($n = 8330$)	TCA users ($n = 8330$)	STD
Renal disease	29 (0.35)	34 (0.41)	0.0098
Any malignancy[c]	166 (1.99)	177 (2.12)	0.0093
Moderate or severe liver disease	25 (0.30)	27 (0.32)	0.0043
Metastatic solid tumor	41 (0.49)	42 (0.50)	0.0017
Psychiatric disease	3836 (46.05)	3844 (46.15)	0.0019
Hypertension	853 (10.24)	933 (11.20)	0.031
Total diabetes mellitus	673 (8.08)	756 (9.08)	0.0356
Chronic pain			
Headache	3618 (43.43)	3386 (40.65)	−0.0564
Low back pain	2189 (26.28)	2270 (27.25)	0.022
Diabetic neuropathic pain	152 (1.82)	181 (2.17)	0.0356
Concomitant medications[a]			
Antiplatelets	149 (1.79)	185 (2.22)	0.0308
Statins	380 (4.56)	447 (5.37)	0.037
Anticoagulants	1 (0.01)	2 (0.02)	0.0089
NSAIDs	5398 (64.80)	5573 (66.90)	0.0443
SNRIs	31 (0.37)	36 (0.43)	0.0095
Opioids	159 (1.91)	180 (2.16)	0.0179
Tramadol or AAP	1553 (18.64)	1642 (19.71)	0.0271
Triptans	128 (1.54)	143 (1.72)	0.0142
Anticonvulsants	752 (9.03)	793 (9.52)	0.017
Number of emergency department visits[a]			0.0218
0	6886 (82.67)	6791 (81.52)	
1–2	1233 (14.80)	1306 (15.68)	
3–5	148 (1.78)	160 (1.92)	
≥ 6	63 (0.76)	73 (0.88)	
Number of surgeries[a]			0.0226
0	6919 (83.06)	6849 (82.22)	
1	1121 (13.46)	1174 (14.09)	
≥ 2	290 (3.48)	307 (3.69)	
Number of hospitalizations[a]			0.0052
0	6930 (83.19)	6836 (82.06)	
1	986 (11.84)	1050 (12.61)	
≥ 2	414 (4.97)	444 (5.33)	

Data are presented as the mean ± standard deviation or *n* (%). *TCAs* tricyclic antidepressants, *STD* standardized difference, *SD* standard deviation, *NSAIDs* non-steroidal anti-inflammatory drugs, *SNRIs* serotonin-norepinephrine reuptake inhibitors, *AAP* acetaminophen

[a]These were assessed in the year prior to cohort entry

[b]Rheumatoid arthritis/collagen vascular diseases

[c]Any malignancy, including lymphoma and leukemia, except malignant neoplasms of the skin

To the best of our knowledge, this is the first study to assess cardiovascular adverse events in patients with chronic pain with low-dose TCA prescriptions. Previous studies on the association between TCAs and cardiovascular adverse events have mostly focused on patients with depression [11, 13] or on those whose indications were not confirmed [12, 32]. TCA treatments for pain have been administered in lower doses than those for depression, as reported in many previous studies and drug information resources [5–7, 18, 36]. Most TCA treatments for pain may be off-label use; however, in Korea, reimbursement is recognized for fibromyalgia, prophylactic migraine (amitriptyline), chronic pain caused by temporomandibular disorders (amitriptyline, nortriptyline), and neuropathic pain (amitriptyline, nortriptyline, and imipramine) [37].

In contrast to the low-dose usage pattern of TCAs in previous studies [13, 14], we divided the TCA user group further into three groups according to the DDD to reflect the

Fig. 2 Covariate balance before and after propensity score matching

Table 2 Proportions of TCAs, amitriptyline, and nortriptyline according to the mean DDD

	Total TCAs	Amitriptyline	Nortriptyline	Others/combined prescriptions
	No. of patients (%)	No. of patients (%)	No. of patients (%)	No. of patients (%)
< 0.15 DDD[a]	5059 (60.73)	4131 (63.82)	881 (56.15)	47 (16.32)
0.15–0.34 DDD	2379 (28.56)	1763 (27.24)	492 (31.36)	124 (43.06)
≥ 0.34 DDD	892 (10.71)	579 (8.94)	196 (12.49)	117 (40.63)
Total	8330	6,473	1,569	288

TCAs tricyclic antidepressants

[a]DDD is expressed as the mean defined daily dose

actual usage pattern. According to the WHO Collaborating Center for Drug Statistics Methodology, the DDD of amitriptyline and nortriptyline is 75 mg/day, but the 10 mg and 25 mg dosage forms are used in South Korea for both drug modes. The dosage forms for amitriptyline and nortriptyline are further divided into 11.25 mg or less, 11.25–25 mg, and 26 mg or more. For imipramine, the DDD is 100 mg/day. The 25 mg dosage form used in South Korea is further divided into 15 mg or less, 15–34 mg, and 34 mg or more. Although previous studies have shown that TCAs are commonly administered at < 0.5 DDD [13, 14], our study demonstrated that more than 60% of TCAs were used at < 0.15 DDD. Based on the safety evaluations of TCAs in previous studies [14, 38], our study shows the same trend as those studies in which low doses were used.

However, a recent study found that the long-term, low-dose use of TCAs was associated with a risk of cardiovascular adverse events [13], which is consistent with our results. Our study further showed that the risk was significantly increased in the low-dose (0.15–0.34 DDD) group. As shown in our analysis of cumulative dosage, a significant increase in the risk of cardiovascular events was noted at a cumulative dose of > 3 DDD and in patients with a cumulative prescription duration of more than 8 days. Thus, although TCAs are used long-term at low doses, the risk of cardiovascular events increases with the cumulative dose. The statistically significant association of the traditional dosing (≥ 0.34 DDD) of TCAs and ischemic stroke found in this study is consistent with the findings of previous studies [39]. However, the increase in risk was not dose-dependent as in previous studies [39, 40].

Nortriptyline is an active metabolite of amitriptyline, which has affinity for serotonin, noradrenalin, α-adrenergics, histamine, muscarinic cholinergics, 5-hydroxytryptamine (HT), N-methyl-D-aspartate, and opioid receptors [41]. Nortriptyline, on the other hand, is more selective for noradrenaline reuptake transporters [41]. Therefore, nortriptyline is generally considered to have less adverse effects, such as

Table 3 Hazard ratio for overall cardiovascular adverse events according to the mean defined daily dose of TCAs, amitriptyline, and nortriptyline

	No. of events	No. of patients	Person-years	Incidence rate (95% CI)[a]	HR (95% CI)
Non-users	244	8330	4024.56	6.06 (5.35–6.87)	Ref
TCAs users	272	8330	4024.32	6.76 (6.00–7.61)	1.12 (0.94–1.33)
< 0.15 DDD[b]	143	5059	2452.12	5.83 (4.95–6.87)	0.96 (0.78–1.18)
0.15–0.34 DDD	95	2379	1142.38	8.32 (6.80–10.17)	1.37 (1.08–1.74)
≥ 0.34 DDD	34	892	429.83	7.91 (5.65–11.07)	1.30 (0.91–1.87)
Amitriptyline					
Non-user	244	8330	4024.56	6.06 (5.35–6.87)	Ref
< 0.15 DDD	121	4131	2000.66	6.05 (5.06–7.23)	1.00 (0.80–1.24)
0.15–0.34 DDD	63	1763	847.35	7.43 (5.81–9.52)	1.23 (0.93–1.62)
≥ 0.34 DDD	25	579	277.41	9.01 (6.09–13.34)	1.48 (0.98–2.24)
Nortriptyline					
Non-user	244	8330	4024.56	6.06 (5.35–6.87)	Ref
< 0.15 DDD	20	881	429.05	4.66 (3.01–7.23)	0.77 (0.49–1.21)
0.15–0.34 DDD	30	492	234.36	12.80 (8.95–18.31)	2.11 (1.44–3.08)
≥ 0.34 DDD	3	196	95.97	3.13 (1.01–9.69)	0.52 (0.17–1.61)

CI confidence interval, *HR* hazard ratio, *TCAs* tricyclic antidepressants, *DDD* defined daily dose

[a]Incidence rates are expressed as events per 100 person-years

[b]DDD is expressed as a mean defined daily dose

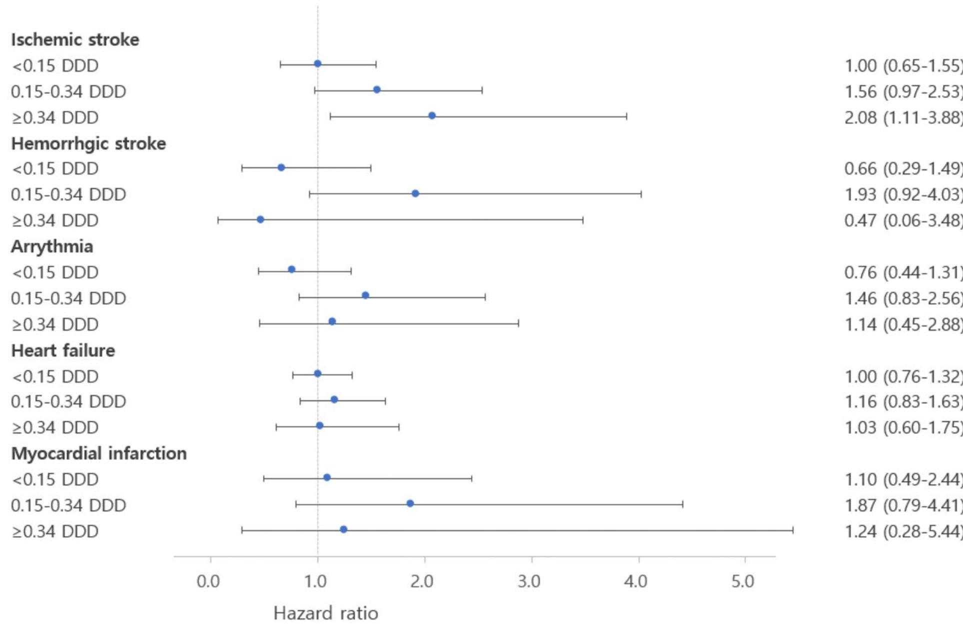


Fig. 3 Hazard ratio of the type of the cardiovascular adverse events according to the mean defined daily dose of tricyclic antidepressants

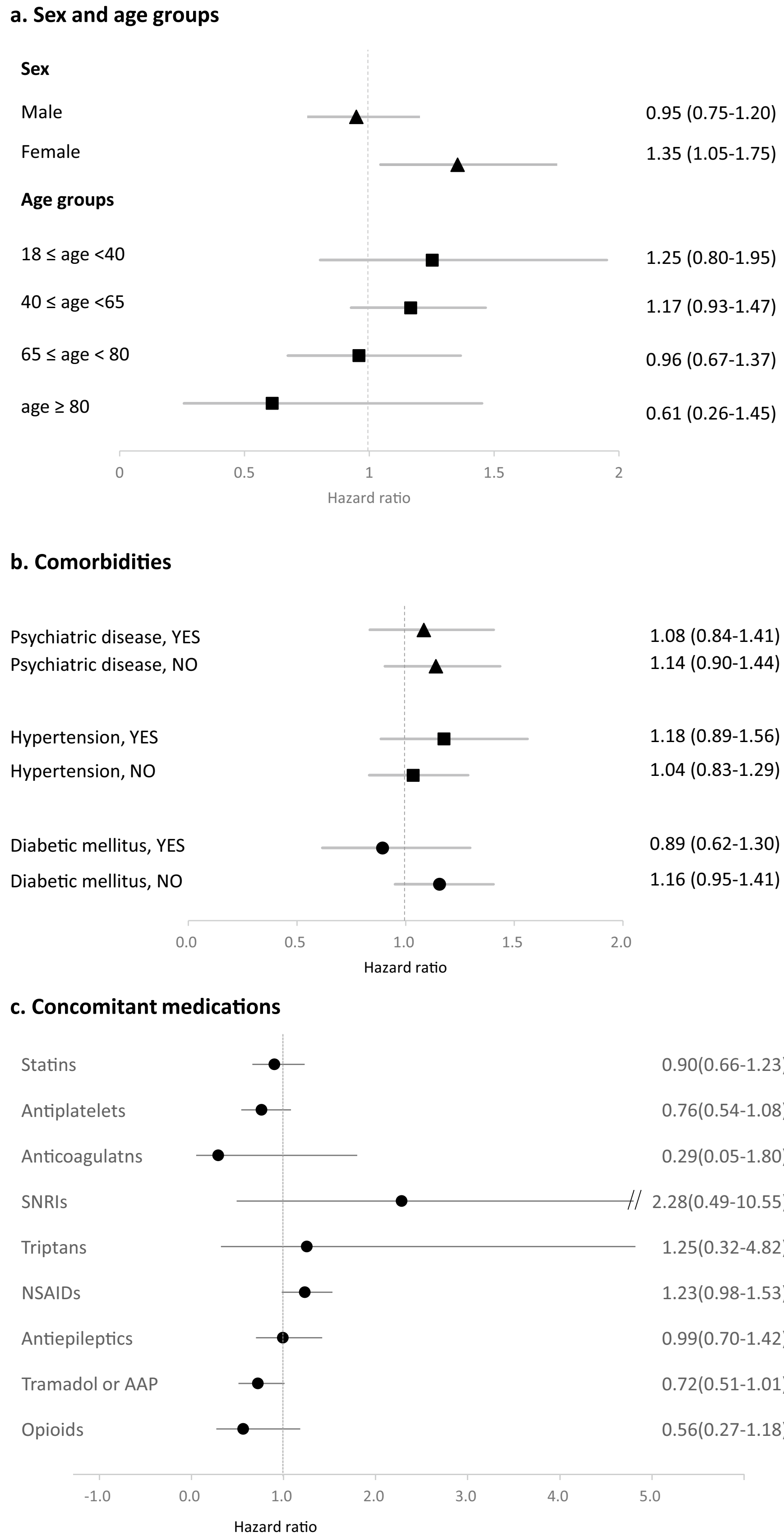
confusion, orthostatic hypotension, and conduction abnormalities, than amitriptyline, and it is sometimes used as a substitute for the latter [42, 43]. However, our results showed that the risk of cardiovascular adverse events was significantly increased at low doses (0.15–0.34 DDD) of nortriptyline compared with amitriptyline. Nortriptyline was also shown to increase the risk of sudden cardiac arrest due to QT prolongation in a previous study [44].

In the subgroup study, female TCA users showed a significantly higher risk of cardiovascular events than female non-users. The risk of cardiovascular events is generally higher in male patients [45]. A study investigating the association of cyclic antidepressants and cardiovascular events [46] found a significantly higher cardiovascular event in male patients, which was inconsistent with our results. However, a study showing that QT prolongation due to the use of TCAs is higher in female than in male patients supports our results [9]. Among concomitant medications, only NSAIDs were associated with a marginal increase in cardiovascular adverse effects. This is in accordance with the results of a previous study [32]; however, when analyzed by dose group, only the low-dose (0.15–0.34 DDD) TCA group showed a significant result, and there was no linear relationship according to the dose (Supplementary Fig. 1). Thus, additional research is needed. NSAIDs themselves have been reported to increase blood pressure and cause cardiovascular adverse effects [47]. However, in our study, we attempted to determine drug-drug interactions between TCAs and NSAIDs by identifying the risk of cardiovascular events in TCAs users and non-users within the NSAID group. However, some NSAIDs may not have been reflected among users of NSAIDs as they are over-the-counter medicines. Although there have been previous studies showing that serotonin and norepinephrine reuptake inhibitors (SNRIs) [48] and triptans [49] have a significant association with cardiovascular events, our study did not show any significant results because of insufficient power.

This study has several strengths. First, this is the first study to evaluate the association between TCA use and cardiovascular events in patients with chronic pain, with a particular interest in the association between antidepressant use and the risk of a cardiovascular event [12–14]. Furthermore, the actual dose usage pattern of TCAs for chronic pain was confirmed by dividing dose usage into three groups. Second, we reduced the confounding factors in baseline characteristics, such as comorbidities and pain severity, using propensity score matching. Third, we attempted to implement the method closest to a randomized controlled trial using the intention-to-treat observational cohort study method [34, 50]. Finally, selection bias was reduced by evaluating a nationwide cohort database, thereby increasing the generalizability of results.

This study has some limitations. First is the lack of validation of diagnosis codes for chronic pain using ICD-10 codes. ICD-10 codes are intended to facilitate billing services and do not reflect the comprehensive judgment of medical professionals. In addition, ICD-10 codes have no direct diagnosis codes for chronic pain. We attempted to define chronic pain comprehensively by combining the results of previous studies that suggested ICD-10 codes for chronic pain conditions by validating ICD-10 for chronic pain [22–25]. Second, although we adjusted for baseline characteristics through propensity score matching, residual confounding factors may still exist. In particular, the atherosclerotic cardiovascular

Fig. 4 Subgroup analysis of hazard ratio for cardiovascular adverse events on patients' sex, age groups, comorbidities (psychiatric disease, hypertension, diabetes mellitus), and concomitants medications (antiplatelets, anticoagulants, statins, NSAIDs, opioids, tramadol or AAP, triptans, SNRIs, antiepileptics). NSAIDs, non-steroidal anti-inflammatory drugs, AAP, acetaminophen, SNRIs, serotonin-norepinephrine reuptake inhibitors



disease risk score [13], lipid parameters, kidney function parameters [31], hemoglobin A1c, and antidiabetics [51] were not considered. However, hypertension, diabetes mellitus, body mass index, smoking habits [52], and exercise habits, which may influence the occurrence of cardiovascular events, were considered. Third, the TCAs included as prescriptions for patients diagnosed with chronic pain were considered to have been used for pain control, but we could not determine whether these treatments were used to induce sleep [53] or for depression without indications of psychiatric diseases. Furthermore, there is a possibility that patients with psychiatric disorders visited medical institutions for reasons other than the purpose of obtaining a TCA prescription. Finally, although the use of TCAs was regarded as prescription use, we could not confirm whether the patients actually used the drugs as prescribed. In addition, since it was not possible to confirm whether patients took over-the-counter NSAIDs, those who did may have been classified into the group that did not take NSAIDs.

Conclusions

Over the first 6 months of treatment with TCAs for chronic pain, we have shown that overall TCAs did not increase cardiovascular adverse events. Nevertheless, low-dose (0.15–0.34 DDD) TCAs and nortriptyline significantly increased cardiovascular adverse events. Furthermore, administering TCAs at the traditional dose (≥ 0.34 DDD) was associated with ischemic stroke. The low-dose use of TCAs may increase the risk of cardiovascular adverse events if the cumulative dose increases with long-term use. Thus, close and long-term monitoring is required for prevention and management.

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Author contribution All authors contributed to the study's conception and design. Material preparation, data collection, and analysis were performed by Sun-Young Jung, Seung Hun You, and Hyunji Koo. The first draft of the manuscript was written by Hyunji Koo. Kyeong Hye Jeong, Nakyung Jeon, Sewon Park, and Sun-Young Jung reviewed and edited previous versions of the manuscript. All authors read and approved the final manuscript.

Availability of data and materials Not applicable.

Declarations

Ethical approval This study protocol was exempted from review by the Institutional Review Board of Chung-Ang University (IRB Number: 1041078–202002-HR-022–01).

Consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

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